Exercise Set 8: Differential Forms and k-Forms

Having built the tangent bundle, we now construct its dual: the cotangent bundle and the algebra of differential forms. A k-form eats k tangent vectors and returns a number, antisymmetrically. The three fundamental operations the exterior derivative d, the interior product ι_X, and the Lie derivative $\mathcal{L}_X$ are related by Cartan's magic formula and together govern how forms behave under flows, symmetries, and integration. The physics payoff is immediate: Maxwell's equations unify into two lines.

Recall of Definitions and Examples

Differential forms

R1. Write the definition of a **k-form** $\omega \in \Omega^k(M)$ as a smooth section of $\Lambda^k T^*M$. What does it mean for ω to be **alternating**? What is $\Omega^0(M)$? What is $\Omega^k(M)$ for $k > \dim M$?

R2. In local coordinates $(x^1, \ldots, x^d)$, write the general local expression for a k-form. What are the component functions? How many independent components does a k-form on $\mathbb{R}^n$ have?

R3. Write the definition of the **wedge product** $\alpha \wedge \beta$ for $\alpha \in \Omega^j(M)$, $\beta \in \Omega^k(M)$. State the **graded commutativity** rule: what is $\alpha \wedge \beta$ in terms of $\beta \wedge \alpha$? What does this imply for $\alpha \wedge \alpha$ when α is a 1-form?

R4. For each of the following, state the degree of the result:

(a) $dx \wedge dy$ on $\mathbb{R}^3$

(b) $f\, dx \wedge dy \wedge dz$ on $\mathbb{R}^3$

(c) $(dx + dy) \wedge dz$ on $\mathbb{R}^3$

(d) $\omega \wedge \eta$ for $\omega \in \Omega^2$, $\eta \in \Omega^3$ on a 6-manifold

(e) $\iota_X \omega$ for $\omega \in \Omega^4$ and $X \in \mathfrak{X}(M)$

The three operations

R5. Write the definition of the **exterior derivative** $d : \Omega^k(M) \to \Omega^{k+1}(M)$. State its three defining properties:

(a) Action on functions ($k = 0$)

(b) Leibniz rule (graded)

(c) Nilpotency

Write the local coordinate formula.

R6. Write the definition of the **interior product** $\iota_X : \Omega^k(M) \to \Omega^{k-1}(M)$. State the graded Leibniz rule for $\iota_X(\alpha \wedge \beta)$. What is $\iota_X \circ \iota_X$?

R7. Write the definition of the **Lie derivative** $\mathcal{L}_X : \Omega^k(M) \to \Omega^k(M)$. State **Cartan's magic formula**: $\mathcal{L}_X = ?$

R8. Complete the following table with the degree of the output and a one-phrase description:

Operation	Input degree	Output degree	What it does
d	k		
ι_X	k		
$\mathcal{L}_X$	k		
$d \circ \iota_X + \iota_X \circ d$	k		

R9. Write the commutation relations:

(a) $d^2 = ?$

(b) $\iota_X^2 = ?$

(c) $[d, \mathcal{L}_X] = ?$

(d) $[\mathcal{L}_X, \mathcal{L}_Y] = ?$

R10. Write the definitions of **closed** and **exact** forms. State the relationship between them (which implies which?). Give one example of each on $\mathbb{R}^2 \setminus \{0\}$.

Geometric interpretation

R11. Describe geometrically what a 1-form, a 2-form, and a 3-form represent in $\mathbb{R}^3$ in terms of level sets, tubes, and boxes. What does $\omega(v_1, \ldots, v_k)$ count?

R12. Fill in the blanks: on $\mathbb{R}^3$, the exterior derivative corresponds to

(a) $d : \Omega^0 \to \Omega^1$ corresponds to _____

(b) $d : \Omega^1 \to \Omega^2$ corresponds to ___

(c) $d : \Omega^2 \to \Omega^3$ corresponds to _____

State the classical identities $\operatorname{curl}(\nabla f) = 0$ and $\operatorname{div}(\operatorname{curl} F) = 0$ as a single equation in the language of forms.

Computation

Exterior derivative d

C1. (Exterior derivative on $\mathbb{R}^3$.) Compute $d\omega$ for each of the following:

(a) $f = x^2y + z$

(b) $\omega = xy\,dx + yz\,dy + xz\,dz$

(c) $\omega = x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy$

(d) $\omega = f(r)\,dx \wedge dy \wedge dz$ where $r = \sqrt{x^2 + y^2 + z^2}$

For (b), identify $d\omega$ as $\mathrm{curl}(\omega)$ written as a 2-form. For (c), identify $d\omega$ as $\mathrm{div}(\omega)$ written as a 3-form.

C2. (Nilpotency check.) For $\omega = xy\,dx + yz\,dy$ on $\mathbb{R}^3$:

(a) Compute $d\omega$.

(b) Compute $d(d\omega)$.

(c) Verify $d^2\omega = 0$ directly the second-order terms cancel due to equality of mixed partials. Show explicitly how the cancellation works.

C3. (Closed but not exact.) The angle form on $\mathbb{R}^2 \setminus \{0\}$ is:

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

(a) Compute $d\omega$. Show it vanishes ω is closed.

(b) In polar coordinates (r, θ), show $\omega = d\theta$.

(c) Why is ω not exact on $\mathbb{R}^2 \setminus \{0\}$?
Hint: compute $\oint_{S^1} \omega$ where S^1 is the unit circle. If $\omega = df$ for some f, this integral would be zero.

(d) On $\mathbb{R}^2$ (no puncture), is ω exact? Why?

C4. (Exterior derivative on S^2.) Using spherical coordinates (θ, φ) on S^2:

(a) Compute $d(\cos\theta)$.

(b) Compute $d(\sin\theta\,d\varphi)$.

(c) The area form on S^2 is $\omega = \sin\theta\,d\theta \wedge d\varphi$. Show $d\omega = 0$. Is ω exact?
Hint: $\int_{S^2} \omega = 4\pi \neq 0$.

(d) Find $\alpha \in \Omega^1(S^2 \setminus \{N, S\})$ such that $d\alpha = \omega$ on the complement of the poles.

Interior product ι_X

C5. (Interior product on $\mathbb{R}^3$.) Let $\Omega = dx \wedge dy \wedge dz$ be the volume form and $X = a\,\partial_x + b\,\partial_y + c\,\partial_z$ a general vector field.

(a) Compute $\iota_X \Omega$. Express the result as a 2-form.

(b) Compute $\iota_{\partial_x}(dx \wedge dy)$, $\iota_{\partial_y}(dx \wedge dy)$, $\iota_{\partial_z}(dx \wedge dy)$.

(c) Let $\omega = dx \wedge dy + dy \wedge dz$. Compute $\iota_X \omega$ for $X = \partial_x + \partial_z$.

(d) Verify nilpotency: compute $\iota_X(\iota_X \Omega)$ for any X.

C6. (Interior product on S^2.) Let $\omega = \sin\theta\, d\theta \wedge d\varphi$ be the area form on S^2.

(a) Compute $\iota_{\partial_\theta}\omega$.

(b) Compute $\iota_{\partial_\varphi}\omega$.

(c) Let $X_3 = \partial_\varphi$ (rotation about the z-axis). Compute $\iota_{X_3}\omega$.

(d) Show that $\iota_{X_3}\omega = -d(\cos\theta)$. This means $\iota_{X_3}\omega$ is exact we will use this in P2.

Lie derivative $\mathcal{L}_X$

C7. (Lie derivative via Cartan's formula.) For each of the following, use $\mathcal{L}_X = d\iota_X + \iota_X d$ to compute $\mathcal{L}_X\omega$. Then verify the result is what you expect from the geometric meaning.

(a) $X = \partial_x$, $\omega = dx$ on $\mathbb{R}^2$.
(Translation preserves dx expect $\mathcal{L}_X\omega = 0$.)

(b) $X = x\partial_x + y\partial_y$ (dilation), $\omega = dx \wedge dy$ on $\mathbb{R}^2$.
(Dilation scales area expect $\mathcal{L}_X\omega = c\,\omega$ for some constant c. What is c?)

(c) $X = -y\partial_x + x\partial_y$ (rotation), $\omega = dx \wedge dy$ on $\mathbb{R}^2$.
(Rotation preserves area expect $\mathcal{L}_X\omega = 0$.)

(d) $X = \partial_\varphi$, $\omega = \sin\theta\, d\theta \wedge d\varphi$ on S^2.
(Rotation about z-axis preserves the area form.)

C8. (All three operations together.) Let $\omega = y\,dx - x\,dy$ on $\mathbb{R}^2$ and $X = x\partial_x + y\partial_y$ (dilation).

(a) Compute $d\omega$.

(b) Compute $\iota_X\omega$.

(c) Compute $d(\iota_X\omega)$.

(d) Compute $\iota_X(d\omega)$.

(e) Compute $\mathcal{L}_X\omega = d(\iota_X\omega) + \iota_X(d\omega)$.

(f) Verify: $\mathcal{L}_X\omega = c\,\omega$ for some c. What is c? Is ω preserved by the dilation flow?

C9. (Pullback of forms.) Let $f : S^1 \to \mathbb{R}^2$ be the inclusion $f(\cos\theta, \sin\theta) = (\cos\theta, \sin\theta)$.

(a) Compute $f^*(dx)$ and $f^*(dy)$. Express the results as multiples of $d\theta$.

(b) Compute $f^*(x\,dy - y\,dx)$.

(c) Compute $f^*(dx \wedge dy)$. Why is the result zero?
(Hint: S^1 is 1-dimensional, so there are no nonzero 2-forms on it.)

(d) Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation by $\pi/2$: $g(x, y) = (-y, x)$. Compute $g^*(dx \wedge dy)$. Is the area form preserved?

Physics: Maxwell's Equations as Differential Forms

Maxwell's equations in vector calculus notation require four separate equations involving $\mathbf{E}$, $\mathbf{B}$, ρ, *and* $\mathbf{J}$. *In the language of differential forms on* $\mathbb{R}^{1,3}$ *(Minkowski spacetime), they collapse into exactly two equations:* $dF = 0$ *and* $d{*F} = J$. *This is not just notational elegance the form language reveals the geometric structure, makes gauge invariance manifest, and generalises immediately to curved spacetime.*

Setup: fields as differential forms

Let (t, x^1, x^2, x^3) be coordinates on $\mathbb{R}^{1,3}$ with the Minkowski metric $\eta = \mathrm{diag}(-1, +1, +1, +1)$. We use the shorthand $x^0 = t$.

The **gauge potential** (or **4-potential**) is the 1-form:

$$A = A_\mu\, dx^\mu = -\phi\, dt + A_1\, dx^1 + A_2\, dx^2 + A_3\, dx^3,$$

where ϕ is the electric potential and $\mathbf{A} = (A_1, A_2, A_3)$ is the magnetic vector potential.

The **field strength tensor** is the 2-form $F = dA$:

$$F = \frac{1}{2} F_{\mu\nu}\, dx^\mu \wedge dx^\nu.$$

The **current** 1-form is:

$$J = J_\mu\, dx^\mu = -\rho\, dt + J_1\, dx^1 + J_2\, dx^2 + J_3\, dx^3,$$

where ρ is the charge density and $\mathbf{J}$ is the current density.

P1. (The field strength tensor.)

(a) Compute $F = dA$ explicitly. The components are:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Write out all the independent components F_{0i} and F_{ij}.

(b) The electric and magnetic fields are defined by:

$$E_i = F_{0i} = \partial_0 A_i - \partial_i A_0, \qquad B_k = \frac{1}{2}\epsilon_{ijk} F_{ij}.$$

Express E_i and B_k in terms of ϕ and $\mathbf{A}$:

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \mathbf{A}.$$

Verify these formulas from $F = dA$.

(c) Write the 2-form F explicitly in terms of $E_1, E_2, E_3, B_1, B_2, B_3$:

$$F = E_1\, dt\wedge dx^1 + E_2\, dt\wedge dx^2 + E_3\, dt\wedge dx^3 + B_3\, dx^1\wedge dx^2 - B_2\, dx^1\wedge dx^3 + B_1\, dx^2\wedge dx^3.$$

Check this is correct by reading off $F_{01} = E_1$, $F_{12} = B_3$, etc.

P2. (The homogeneous Maxwell equations: $dF = 0$.)

(a) Since $F = dA$, compute $dF = d(dA)$. Why is $dF = 0$ an automatic consequence of $F = dA$?

(b) Compute $dF = 0$ explicitly by expanding in components. Show that the equation $dF = 0$ gives exactly two of the four Maxwell equations:

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0.$$

Hint: collect the $dx^i \wedge dx^j \wedge dx^k$ components to get $\nabla \cdot \mathbf{B} = 0$, and the $dt \wedge dx^i \wedge dx^j$ components to get Faraday's law.

(c) These are the **source-free** equations they don't involve the charge ρ or current $\mathbf{J}$. Why does $dF = 0$ follow automatically from $F = dA$? What does this say about gauge invariance?

(d) **Gauge invariance.** Under the gauge transformation $A \to A + d\lambda$ for any smooth function λ: show that $F = dA$ is unchanged. This is why F, not A, is the physically observable field.

P3. (The inhomogeneous Maxwell equations: $d{*}F = J$.)

The Hodge star $* : \Omega^k \to \Omega^{4-k}$ on Minkowski space maps 2-forms to 2-forms. Its action on the basis is:

$$*(dt \wedge dx^i) = \epsilon_{ijk}\, dx^j \wedge dx^k$$
$$*(dx^i \wedge dx^j) = -\epsilon_{ijk}\, dt \wedge dx^k$$

(the signs come from the Lorentzian signature $(-,+,+,+)$).

(a) Compute $*F$. Express it in terms of $\mathbf{E}$ and $\mathbf{B}$:

$$*F = B_1\, dt\wedge dx^1 + B_2\, dt\wedge dx^2 + B_3\, dt\wedge dx^3 - E_3\, dx^1\wedge dx^2 + E_2\, dx^1\wedge dx^3 - E_1\, dx^2\wedge dx^3.$$

Notice: $*F$ is obtained from F by $\mathbf{E} \to \mathbf{B}$, $\mathbf{B} \to -\mathbf{E}$ (electromagnetic duality).

(b) Compute $d(*F)$ and show it gives the other two Maxwell equations:

$$\nabla \cdot \mathbf{E} = \rho, \qquad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J}.$$

by comparing $d(*F) = *J$ component by component.

(c) The full Maxwell equations in differential form language are:

$$\boxed{dF = 0, \qquad d*F = J.}$$

Write out all four classical Maxwell equations from memory, then verify each one follows from these two.

P4. (Charge conservation from $d^2 = 0$.)

The conservation of electric charge $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$ follows automatically from the Maxwell equations.

(a) Apply d to both sides of $d*F = J$. What do you get on the left using $d^2 = 0$? Conclude $dJ = 0$.

(b) Expand $dJ = 0$ in components. Show this is exactly the charge conservation equation $\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$.

(c) This shows that charge conservation is not an independent law it is a **mathematical consequence** of $d^2 = 0$ applied to the Maxwell equations. Write one sentence explaining what this means physically.

P5. (The action principle.)

The Maxwell action is:

$$S[A] = -\frac{1}{2} \int_{\mathbb{R}^{1,3}} F \wedge *F - \int_{\mathbb{R}^{1,3}} A \wedge *J.$$

(a) The first term $F \wedge *F = \frac{1}{2}(|\mathbf{B}|^2 - |\mathbf{E}|^2)\, dt\, dx^1\, dx^2\, dx^3$ is the electromagnetic Lagrangian density. Verify this for the case $\mathbf{B} = 0$: $F \wedge *F = -|\mathbf{E}|^2\, dt\, dx^1\, dx^2\, dx^3$.

(b) The equation of motion $\delta S / \delta A = 0$ gives $d * F = J$. The equation $dF = 0$ follows automatically from $F = dA$. So the action principle already encodes all four Maxwell equations. Why is this more efficient than varying $\mathbf{E}$ and $\mathbf{B}$ separately?

(c) Under $A \to A + d\lambda$, show that $S[A]$ is invariant (assuming J satisfies $dJ = 0$). *Hint: $F \to F$, and use integration by parts for the second term.*

Everything the equations, the conservation law, the gauge symmetry flows from the single identity $d^2 = 0$. This is the power of the differential forms language: what looks

like four separate laws in vector calculus is a single geometric statement about the exterior derivative of a 2-form.

The next step which we have not yet taken is to put a metric on the manifold, which is needed to define the Hodge star $*$ and the action. The Maxwell equations $dF = 0$ make sense on any smooth manifold. The sourced equation $d * F = J$ requires a metric. This is the distinction between topology (which $dF = 0$ sees) and geometry (which $d * F = J$ requires).***Reflection.*** *Maxwell's equations in differential form language are $dF = 0$ and $d*F = J$. The first is automatic from $F = dA$ and $d^2 = 0$. The second gives the sourced equations. Charge conservation $dJ = 0$ follows from $d^2 = 0$ applied to the second equation. Gauge invariance $A \to A + d\lambda$ is manifest because $d^2\lambda = 0$.*

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